MATH 241: Calculus III

Fall 2026

Lecture 1: Calculus Review & 3D Coordinate Systems

Background & Motivation

Welcome to Calculus III. Everything you learned in Calc I and II—limits, derivatives, integrals, Taylor series—was about functions of a single variable. This semester, we generalize all of it to functions of two, three, or more variables. Today we start with a rapid review of the tools you'll need, then introduce the 3D coordinate systems that let us describe points in space. By the end of the semester, you'll be integrating over surfaces and applying theorems that unify all of calculus.

1. Core Concepts

- Limit: $\lim_{x \to a} f(x) = L$ means ____________________________________
- Derivative: $f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$
- L'Hôpital's Rule: For $\frac{0}{0}$ or $\frac{\infty}{\infty}$: $\lim_{x \to a} \frac{f(x)}{g(x)} =$ __________________

2. Essential Rules

Derivatives and Integrals

Derivatives:

$$(uv)' = \underline{\hspace{4cm}}$$
$$\left(\frac{u}{v}\right)' = \underline{\hspace{4cm}}$$
$$\frac{d}{dx}[f(g(x))] = \underline{\hspace{4cm}}$$
$$\frac{d}{dx}[x^n] = \underline{\hspace{2.5cm}}$$
$$\frac{d}{dx}[\ln(x)] = \underline{\hspace{2.5cm}}$$
$$\frac{d}{dx}[e^x] = \underline{\hspace{2.5cm}}$$

Integrals:

$$\int_a^b f(x)\,dx = \underline{\hspace{4cm}}$$
$$\int u\,dv = \underline{\hspace{4cm}}$$
$$\int x^n\,dx = \underline{\hspace{4cm}}$$
$$\int \frac{1}{x}\,dx = \underline{\hspace{4cm}}$$
$$\int e^x\,dx = \underline{\hspace{4cm}}$$
$$\int \sin(x)\,dx = \underline{\hspace{4cm}}$$

3. Taylor Series

– General form around $x = a$:

$$f(x) = \underline{\hspace{6cm}}$$

– Common series:

$$e^x = \underline{\hspace{8cm}}$$
$$\sin(x) = \underline{\hspace{8cm}}$$

4. 3D Coordinate Systems

Rectangular Coordinates

– Point in 3D: (x, y, z) means ______________________________

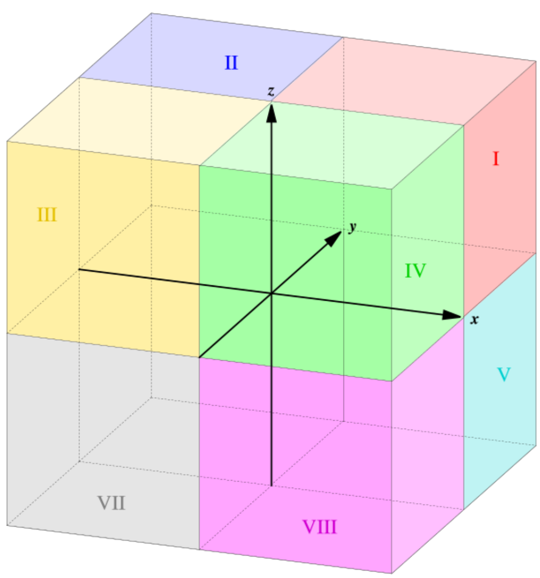
Distance and Midpoint in 3D

Distance between (x_1, y_1, z_1) and (x_2, y_2, z_2):

$$d = \underline{\hspace{6cm}}$$

Midpoint:

$$M = \underline{\hspace{6cm}}$$



The eight octants of 3D space.

– Coordinate planes: xy-plane: ______________ xz-plane: ______________ yz-plane: ______________

Alternative Coordinate Systems

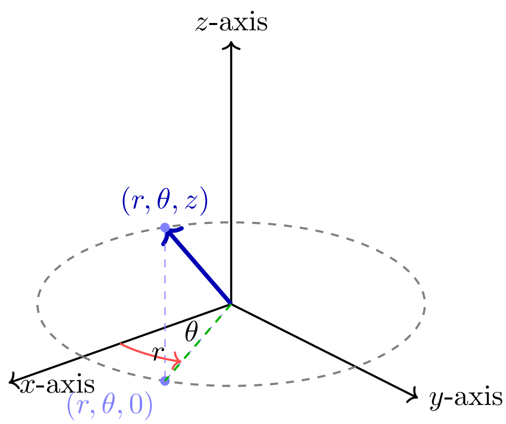
Polar and Cylindrical

Polar (r, θ):

- r: ________________
- θ: ________________
- To rectangular:
  $x = $ ________________, $y = $ ________________
- From rectangular:
  $r = $ ________________, $\theta = $ ________________

Cylindrical (r, θ, z):

- r: ________________
- θ: ________________
- z: ___________
- To rectangular:
  $x = $ ________________, $y = $ ________________, $z = $ ________________



A point in cylindrical coordinates (r, θ, z): radius r from the z-axis, angle θ in the xy-plane, and vertical height z.

Spherical Coordinates (ρ, θ, ϕ)

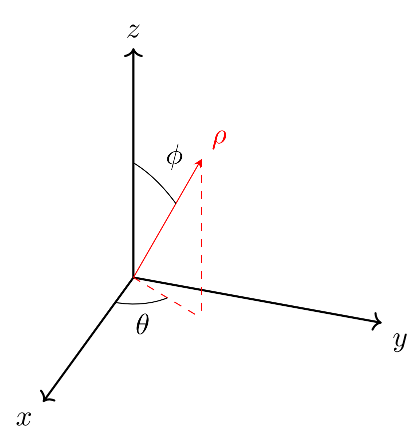
- ρ: ________________
- θ: ______________________
- ϕ: ______________________

– To rectangular:

$$x = \underline{\hspace{4cm}}, \quad y = \underline{\hspace{4cm}}, \quad z = \underline{\hspace{4cm}}$$

– From rectangular:

$$\rho = \underline{\hspace{4cm}}, \quad \theta = \underline{\hspace{4cm}}, \quad \phi = \underline{\hspace{4cm}}$$



A point in spherical coordinates (ρ, θ, ϕ): distance ρ from the origin, azimuthal angle θ in the xy-plane, and polar angle ϕ measured from the $+z$-axis.

Practice Problems

Problem 1. Evaluate $\lim_{x \to 0} \dfrac{e^x - 1 - x - \frac{x^2}{2}}{x^3}$.

Problem 2. Find first three terms of Taylor series for $\ln(1 + x)$ at $a = 0$.

Problem 3. Convert cylindrical point $(3, \pi/4, 2)$ to rectangular coordinates.

Problem 4. Find distance between points $(1, 1, 1)$ and $(4, 5, 2)$.

Problem 5. Convert rectangular point $(0, 2, 2)$ to spherical coordinates.